

Scientific Calculator

```
#include <iostream>

#include <cmath>

using namespace std;

int main()
{
    int choice;

    double a, b;

    cout<<"cu26250157\nKrish Choudhary\n";

    do
    {
        cout << "\nSCIENTIFIC CALCULATOR\n";
        cout << "1. Addition\n";
        cout << "2. Subtraction\n";
        cout << "3. Multiplication\n";
        cout << "4. Division\n";
        cout << "5. Modulus\n";
        cout << "6. Power\n";
        cout << "7. Square\n";
        cout << "8. Cube\n";
        cout << "9. Square Root\n";
        cout << "10. Absolute Value\n";
        cout << "11. Sin\n";
        cout << "12. Cos\n";
        cout << "13. Tan\n";
        cout << "14. Cot\n";
```

```
cout << "15. Sec\n";
```

```
cout << "16. Cosec\n";
```

```
cout << "0. Exit\n";
```

```
cout << "\nEnter your choice: ";
```

```
cin >> choice;
```

```
switch(choice)
```

```
{
```

```
    case 1:
```

```
        cout << "Enter two numbers: ";
```

```
        cin >> a >> b;
```

```
        cout << "Addition = " << a + b << endl;
```

```
        break;
```

```
    case 2:
```

```
        cout << "Enter two numbers: ";
```

```
        cin >> a >> b;
```

```
        cout << "Subtraction = " << a - b << endl;
```

```
        break;
```

```
    case 3:
```

```
        cout << "Enter two numbers: ";
```

```
        cin >> a >> b;
```

```
        cout << "Multiplication = " << a * b << endl;
```

```
        break;
```

```
    case 4:
```

```
        cout << "Enter two numbers: ";
```

```
cin >> a >> b;
```

```
if(b != 0)
```

```
    cout << "Division = " << a / b << endl;
```

```
else
```

```
    cout << "Division by zero is not possible." << endl;
```

```
break;
```

```
case 5:
```

```
    int x, y;
```

```
    cout << "Enter two integers: ";
```

```
    cin >> x >> y;
```

```
    if(y != 0)
```

```
        cout << "Modulus = " << x % y << endl;
```

```
    else
```

```
        cout << "Modulus by zero is not possible." << endl;
```

```
    break;
```

```
case 6:
```

```
    cout << "Enter base: ";
```

```
    cin >> a;
```

```
    cout << "Enter power: ";
```

```
    cin >> b;
```

```
    cout << "Power = " << pow(a, b) << endl;
```

```
    break;
```

case 7:

```
cout << "Enter a number: ";
```

```
cin >> a;
```

```
cout << "Square = " << a * a << endl;
```

```
break;
```

case 8:

```
cout << "Enter a number: ";
```

```
cin >> a;
```

```
cout << "Cube = " << a * a * a << endl;
```

```
break;
```

case 9:

```
cout << "Enter a number: ";
```

```
cin >> a;
```

```
if(a >= 0)
```

```
    cout << "Square Root = " << sqrt(a) << endl;
```

```
else
```

```
    cout << "Invalid number." << endl;
```

```
break;
```

case 10:

```
cout << "Enter a number: ";
```

```
cin >> a;
```

```
cout << "Absolute Value = " << fabs(a) << endl;  
break;
```

case 11:

```
cout << "Enter angle in degrees: ";  
cin >> a;
```

```
a = a * 3.14159 / 180;  
cout << "Sin = " << sin(a) << endl;  
break;
```

case 12:

```
cout << "Enter angle in degrees: ";  
cin >> a;
```

```
a = a * 3.14159 / 180;  
cout << "Cos = " << cos(a) << endl;  
break;
```

case 13:

```
cout << "Enter angle in degrees: ";  
cin >> a;
```

```
a = a * 3.14159 / 180;
```

```
if(cos(a) != 0)  
    cout << "Tan = " << tan(a) << endl;  
else  
    cout << "Tan is not defined." << endl;
```

```
break;
```

case 14:

```
cout << "Enter angle in degrees: ";
```

```
cin >> a;
```

```
a = a * 3.14159 / 180;
```

```
if(sin(a) != 0)
```

```
    cout << "Cot = " << 1 / tan(a) << endl;
```

```
else
```

```
    cout << "Cot is not defined." << endl;
```

```
break;
```

case 15:

```
cout << "Enter angle in degrees: ";
```

```
cin >> a;
```

```
a = a * 3.14159 / 180;
```

```
if(cos(a) != 0)
```

```
    cout << "Sec = " << 1 / cos(a) << endl;
```

```
else
```

```
    cout << "Sec is not defined." << endl;
```

```
break;
```

case 16:

```
cout << "Enter angle in degrees: ";
```

```
cin >> a;
```

```
a = a * 3.14159 / 180;
```

```
if(sin(a) != 0)
```

```
    cout << "Cosec = " << 1 / sin(a) << endl;
```

```
else
```

```
    cout << "Cosec is not defined." << endl;
```

```
break;
```

```
case 0:
```

```
    cout << "\nCalculator closed. Thank you!" << endl;
```

```
break;
```

```
default:
```

```
    cout << "\nInvalid choice! Please try again." << endl;
```

```
}
```

```
} while(choice != 0);
```

```
return 0;
```

```
}
```